Coherence-Bounded Operational Locality: An Experimental Test of Simultaneous Phase Synchronization vs. Bell Correlations

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November 2026

Abstract

Bell's theorem (1964) demonstrates that no physical theory grounded in classical local hidden variables can reproduce all statistical predictions of quantum mechanics. Within the philosophy of physics and quantum foundations, this result is frequently interpreted as requiring a fundamental tension between quantum entanglement and relativistic locality. In this paper, we argue that Bell's theorem is completely vindicated in its mathematical and empirical scope, and that the apparent tension arises from conflating quantum correlations with an idealized, infinite-precision *naive classical point-locality*. By extending the Operational Simultaneity Horizon ($\mathcal{L}_{\text{sim}}$) formalism developed in earlier installments of this series, we introduce **Coherence-Bounded Operational Locality**. We prove that while post-selected Bell–CHSH correlations depend solely on local phase coherence $\Phi_\tau(\Omega)$ independent of spatial separation, active simultaneous phase synchronization across a physical channel is strictly bounded by the round-trip signal propagation time $\Delta\tau_{\text{sig}}$ relative to the system's phase decoherence rate Γ_{tot}. We derive a dimensionless **Spatial Locality Decay Factor** ($\mathcal{D}_{\text{loc}}$) that cleanly separates passive non-local quantum correlations from active operational causal control. Finally, we propose a concrete experimental protocol—the **Delayed-Feedback Entangled Optical Clock Experiment**—spanning optical fiber baselines from $\Delta\ell = 10\,\text{m}$ to 50 km under tunable environmental noise Γ_{tot} to verify this divergence empirically.

1 Introduction: Re-examining Locality After Bell

Since John Stewart Bell's landmark 1964 theorem and its subsequent loophole-free experimental confirmations, debates over the foundations of quantum mechanics have centered on the compatibility of quantum non-locality with relativistic spacetime structure. In standard four-dimensionalist or Block Universe accounts of relativity, spacetime events are fixed within a Lorentzian manifold where causal influence is strictly restricted to the light cone. Against this backdrop, the violation of the Clauser–Horne–Shimony–Holt (CHSH) inequality ($S_{\text{CHSH}} > 2$) across spacelike-separated wings is often framed as "spooky action at a distance" or a breakdown of spatial locality.

However, a rigorous distinction must be drawn between two physically distinct phenomena:

1. **Correlation across spacelike intervals:** Statistical dependencies revealed through post-selected classical coincidence records.

2. **Operational causal synchronization:** The physical establishment, verification, and maintenance of a shared phase reference between distant local laboratories in real time.

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Bell's theorem definitively rules out local hidden-variable models that assume classical point-like properties exist independently of measurement contexts. It does not, however, imply that physical observers can transmit actionable phase information superluminally, nor does it render spatial separation physically irrelevant to metrological coordination.

What requires refinement is not Bell's theorem, which remains completely vindicated, but rather the unexamined classical assumption of **naive point-locality**—the idealization that physical systems possess infinitely sharp, zero-latency coordinate states independent of the finite thermodynamic and electromagnetic channels required to synchronize them.

2 Theoretical Framework: Coherence-Bounded Operational Locality

To formalize the boundary between passive quantum correlation and active relativistic synchronization, we evaluate a bipartite entangled optical clock network separated by a proper physical baseline $\Delta\ell$ within a refractive medium of effective group index N_{eff} (such as a telecom optical fiber, where $N_{\text{eff}} \approx 1.468$).

2.1 Round-Trip Signal Delay and the Operational Simultaneity Horizon

Establishing operational simultaneity between two spatially separated nodes A and B requires a closed two-way electromagnetic exchange (Einstein–Poincaré synchronization).

Definition 1 (Round-Trip Signal Delay)**.** *The minimum round-trip signal delay $\Delta\tau_{\text{sig}}$ across proper spatial baseline $\Delta\ell$ in a waveguide or medium of effective group refractive index N_{eff} is governed by*

$$\Delta\tau_{\text{sig}} = \frac{2\Delta\ell}{cN_{\text{eff}}}, \tag{1}$$

where c is the speed of light in vacuum.

During this round-trip interval $\Delta\tau_{\text{sig}}$, both local oscillators and the quantum channel are subject to total phase decoherence at rate Γ_{tot} (measured in s^{-1}), encompassing thermal drift, acoustic phase noise, and quantum projection noise. Let $\rho_\tau \in (0, 1]$ denote the dimensionless target phase-locking threshold parameter required to sustain a valid operational simultaneity frame.

As established in Paper 9 (*Operational Simultaneity Horizons and the Resolution of the Andromeda Paradox*), active phase synchronization remains operationally valid only when the round-trip signal delay does not exceed the coherence window $\tau_{\text{coh}} = \rho_\tau/\Gamma_{\text{tot}}$. Setting $\Delta\tau_{\text{sig}} = \tau_{\text{coh}}$ yields the critical spatial scale.

Definition 2 (Operational Simultaneity Horizon)**.** *The Operational Simultaneity Horizon $\mathcal{L}_{\text{sim}}$ is the maximum spatial baseline across which closed-loop phase synchronization can be sustained against total decoherence rate Γ_{tot} at target threshold ρ_τ:*

$$\mathcal{L}_{\text{sim}} = \frac{cN_{\text{eff}}\rho_\tau}{2\Gamma_{\text{tot}}}. \tag{2}$$

Beyond $\mathcal{L}_{\text{sim}}$, closed-loop phase feedback fails to converge faster than stochastic phase diffusion erases the shared reference frame.

2.2 Operational Simultaneity Fidelity vs. Post-Selected CHSH Correlations

Consider an entangled photon-pair source distributing time-bin or polarization-entangled states correlated with local optical clock transitions driven at coupling parameter Ω. Let $\Phi_\tau(\Omega) \in [0, 1]$

represent the intrinsic local temporal coherence function of the entangled clock states prior to spatial channel transit.

When node A and node B perform local projective measurements and compare timestamps *post-hoc* via classical channels, the resulting post-selected CHSH Bell correlation depends purely on the preserved state purity $\Phi_\tau(\Omega)$:

$$S_{\mathrm{CHSH}}(\Omega) = 2\sqrt{2} \cdot \Phi_\tau(\Omega). \tag{3}$$

Provided $\Phi_\tau(\Omega) > 1/\sqrt{2} \approx 0.7071$, the system violates the classical Bell bound ($S_{\mathrm{CHSH}} > 2$) regardless of the physical baseline $\Delta\ell$, assuming dispersion compensation and low photon loss.

In stark contrast, if nodes A and B attempt to use the channel for **real-time, active operational phase synchronization** without post-selection, the actionable **Operational Simultaneity Fidelity** ($\mathcal{S}_{\mathrm{sim}}$) decays exponentially with the ratio of the physical baseline to the Operational Simultaneity Horizon:

$$\mathcal{S}_{\mathrm{sim}}(\Omega, \Delta\ell) = \Phi_\tau(\Omega) \cdot \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\mathrm{sim}}}\right). \tag{4}$$

2.3 The Spatial Locality Decay Factor ($\mathcal{D}_{\mathrm{loc}}$)

Theorem 1 (Coherence-Bounded Operational Locality)**.** *The normalized ratio between active Operational Simultaneity Fidelity $\mathcal{S}_{\mathrm{sim}}(\Omega, \Delta\ell)$ and post-selected quantum Bell correlation $S_{\mathrm{CHSH}}(\Omega)$ defines the dimensionless* ***Spatial Locality Decay Factor*** $\mathcal{D}_{\mathrm{loc}} \in (0, 1]$*:*

$$\mathcal{D}_{\mathrm{loc}} = \frac{2\sqrt{2} \cdot \mathcal{S}_{\mathrm{sim}}(\Omega, \Delta\ell)}{S_{\mathrm{CHSH}}(\Omega)} = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\mathrm{sim}}}\right) = \exp\left(-\frac{2\Gamma_{\mathrm{tot}}\Delta\ell}{cN_{\mathrm{eff}}\rho_\tau}\right). \tag{5}$$

Consequently, active operational phase control vanishes exponentially for $\Delta\ell \gg \mathcal{L}_{\mathrm{sim}}$ even when $S_{\mathrm{CHSH}}(\Omega) > 2$ remains maximally violated.

Proof. Substituting Eqs. (3) and (4) into the normalized ratio yields

$$\frac{2\sqrt{2} \cdot \Phi_\tau(\Omega) \exp(-\Delta\ell/\mathcal{L}_{\mathrm{sim}})}{2\sqrt{2} \cdot \Phi_\tau(\Omega)} = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\mathrm{sim}}}\right),$$

for all $\Phi_\tau(\Omega) > 0$. Substituting $\mathcal{L}_{\mathrm{sim}}$ from Eq. (2) gives the explicit dependence on $\Delta\ell$, Γ_{tot}, N_{eff}, and ρ_τ. □

Table 1 delineates the three physical regimes governed by $\mathcal{D}_{\mathrm{loc}}$.

3 Experimental Proposal: Delayed-Feedback Entangled Optical Clock Test

To test Coherence-Bounded Operational Locality directly in the laboratory, we propose the **Delayed-Feedback Entangled Optical Clock Experiment**.

3.1 Apparatus Architecture

1. **Entangled Optical Clock Nodes (A and B):** Two trapped-ion or optical-lattice clock ensembles (^{87}Sr or ^{171}Yb$^+$) prepared in a shared Bell state via remote photon-mediated entanglement, maintaining high intrinsic coherence $\Phi_\tau(\Omega) \geq 0.92$.

2. **Variable Optical Fiber Baseline ($\Delta\ell$):** Spools of dispersion-shifted telecom fiber ($N_{\mathrm{eff}} = 1.468$) allowing discrete physical path lengths of $\Delta\ell \in \{10\,\mathrm{m}, 100\,\mathrm{m}, 1\,\mathrm{km}, 10\,\mathrm{km}, 50\,\mathrm{km}\}$.

Table 1: Physical and philosophical regimes of Coherence-Bounded Operational Locality.

Regime	**Baseline**	$\mathcal{D}_{\text{loc}}$ **Value**	**Physical & Philosophical Interpretation**
Near-Field Coherent	$\Delta\ell \ll \mathcal{L}_{\text{sim}}$	$\mathcal{D}_{\text{loc}} \to 1$	Naive classical point-locality holds as an accurate engineering approximation; active phase locking and Bell correlations coincide.
Horizon Transition	$\Delta\ell \approx \mathcal{L}_{\text{sim}}$	$\mathcal{D}_{\text{loc}} = e^{-1} \approx 0.368$	Operational simultaneity degrades rapidly while post-selected Bell violations ($S_{\text{CHSH}} > 2$) remain maximal.
Far-Field Decoupled	$\Delta\ell \gg \mathcal{L}_{\text{sim}}$	$\mathcal{D}_{\text{loc}} \to 0$	Active real-time synchronization is impossible; quantum entanglement persists solely as a non-signaling, post-selected correlation within the Block Universe.

3. **Tunable Noise Injection (Γ_{tot}):** Controlled phase-modulating electro-optic noise generators coupled to the local clock lasers, allowing Γ_{tot} to be swept continuously from $10^2\,\text{s}^{-1}$ (ultra-stable cryogenic baseline) to $10^5\,\text{s}^{-1}$ (high-decoherence regime).

4. **Dual Readout Channels:**

   - *Channel 1 (Post-Selected CHSH):* Time-tagged single-photon detectors recording coincidence events across randomized measurement bases to compute $S_{\text{CHSH}}(\Omega)$.
   - *Channel 2 (Closed-Loop Operational Sync):* Real-time phase-error feedback loop attempting to lock Clock B to Clock A across $\Delta\ell$ without post-selection, directly measuring $\mathcal{S}_{\text{sim}}(\Omega, \Delta\ell)$.

3.2 Quantitative Predictions

For a representative high-precision setting ($\rho_\tau = 0.5$, $N_{\text{eff}} = 1.468$, $\Gamma_{\text{tot}} = 5.0 \times 10^3\,\text{s}^{-1}$), the Operational Simultaneity Horizon evaluates to:

$$\mathcal{L}_{\text{sim}} = \frac{(2.9979 \times 10^8\,\text{m/s})(1.468)(0.5)}{2(5.0 \times 10^3\,\text{s}^{-1})} \approx 22.0\,\text{km}. \tag{6}$$

Across the variable fiber baselines, the experiment predicts a sharp bifurcation between S_{CHSH} and $\mathcal{D}_{\text{loc}}$:

- **At $\Delta\ell = 10\,\text{m}$:** $\mathcal{D}_{\text{loc}} \approx 0.9995$. Both $S_{\text{CHSH}} \approx 2.60$ and $\mathcal{S}_{\text{sim}} \approx 0.92$ exhibit near-ideal performance.
- **At $\Delta\ell = 10\,\text{km}$:** $\mathcal{D}_{\text{loc}} \approx 0.635$. S_{CHSH} remains at ≈ 2.60, while $\mathcal{S}_{\text{sim}}$ drops to ≈ 0.58.
- **At $\Delta\ell = 50\,\text{km}$:** $\mathcal{D}_{\text{loc}} \approx 0.103$. S_{CHSH} remains robustly above the classical limit ($S_{\text{CHSH}} \approx 2.60 > 2$), while real-time operational simultaneity collapses ($\mathcal{S}_{\text{sim}} \approx 0.095$).

4 Philosophical Implications for Spacetime and Locality

Within a neutral four-dimensionalist ontology, all measurement events at A and B occupy fixed coordinates on the spacetime manifold. Bell's theorem proves that the joint probability

distribution of these events cannot be factorized into local classical hidden states assigned to individual worldlines.

Coherence-Bounded Operational Locality demonstrates why this non-factorizability never threatens relativistic causality. Physical locality is not an abstract geometric postulate of zero-volume mathematical points; it is an *operational capacity* governed by finite signal propagation ($\Delta\tau_{\text{sig}}$) and thermodynamic stability (Γ_{tot}). By replacing naive classical point-locality with $\mathcal{D}_{\text{loc}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$, quantum mechanics and special relativity coexist without contradiction: entanglement reflects global structural consistency across spacetime, while operational synchronization remains strictly bounded by the light cone and the arrow of decoherence.

5 Conclusion

Bell's theorem stands completely vindicated as a foundational pillar of modern physics. What collapses under experimental scrutiny is not relativistic locality, but the classical idealization of instantaneous, zero-noise point-locality. Through the Operational Simultaneity Horizon $\mathcal{L}_{\text{sim}}$ and the Spatial Locality Decay Factor $\mathcal{D}_{\text{loc}}$, we establish a testable, quantitative bridge between quantum information theory and relativistic metrology.

AI Collaboration & Authorship Transparency Statement

This work is the result of human–AI scholarly collaboration. The core philosophical framework, conceptual architecture, operational definitions, and theoretical direction were conceived and authored by **Luís Pimenta Damásio** (ORCID: 0009-0001-7606-7557). Mathematical formalization, analytical verification, and LaTeX typesetting support were provided by **Gemini** (Google AI) under the direct supervision, review, and editorial authority of the human author.

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